

Monte Carlo Simulation

Ada Lovelace Team

Instructor Name: Dr Maha Amin

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Team Members:

<i>Name:</i>	<i>Code:</i>
Nancy Ashraf George	91240807
Habiba Khaled Abdelsalam	91240255
Maryam Refaat Abdelhamid	91240738
Ebraam Ashraf	91240051
Muhammad Khaled	91240653

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Abstract

This study statistically investigates the relationship between Monte Carlo sampling parameters (Sample Count N and Sampling Strategy) and the resulting image quality and computational performance in ray-traced photorealistic rendering. As modern graphics demand both high realism and efficiency, the objective was to move from qualitative observations of rendering noise to a data-driven framework for optimizing the stochastic rendering process.

The experiment utilized a custom path tracer, varying the **Sample Count N in {1, 2, ..., 4096}** and comparing **Uniform Sampling** against **Importance Sampling** across a scene designed for complex global illumination. Key performance indicators—Rendering Time (**T**), Mean Squared Error (**MSE**), and Peak Signal-to-Noise Ratio (**PSNR**)—were recorded for each run.

1. Introduction

1.1 Background and Significance:

- As modern films, animations, and games demand higher photorealism, rendering algorithms must efficiently balance visual quality with performance. Rendering realistic 3D scenes is computationally challenging, and traditional deterministic ray tracing struggles to model complex effects like global illumination and soft shadows. Monte Carlo simulation addresses these limitations by estimating the Rendering Equation through random sampling. This study examines how different sampling strategies and sample counts affect image quality, rendering time, and overall realism, aiming to optimize the trade-off between accuracy and computational cost.
- The Rendering Equation (Kajiya, 1986) mathematically defines light transport in a scene. Since it's impossible to solve analytically, Monte Carlo integration is used to approximate it via random sampling.
- Each pixel's colour becomes a random variable whose expected value approximates the true illumination.
- The experiment focuses on how increasing the **number of samples (N)** reduces noise (variance) in the rendered image and how **sampling method** influences convergence speed and realism.

1.2 Research Objectives:

The goal is to quantify and statistically validate the relationship between Monte Carlo sampling parameters (Sample Count and Sampling Strategy) and the resulting image quality and computational performance in ray tracing, with the objective of identifying the most efficient rendering approach

->Here is the goal broken down by its three core components:

1. Primary Objective (Quantification)

- To precisely model the scaling behavior of computational cost and image quality metrics (PSNR, RMSE) as the Sample Count (N) is increased, verifying the theoretical $O(1/\sqrt{N})$ convergence rate for noise reduction and the expected linear scaling of rendering time ($T \propto N$).

2. Comparative Objective (Validation)

- To statistically determine, using advanced hypothesis testing (t-tests, ANOVA, Multiple Regression), whether Importance Sampling yields a significantly higher mean image quality (PSNR) and a more efficient quality-to-cost ratio compared to **Uniform Sampling** under equivalent computational conditions.

3. Practical Objective (Optimization)

- To identify the point of diminishing returns—the sample count threshold where the marginal gain in image quality no longer justifies the linear increase in computational cost—thereby providing actionable data for setting optimal rendering budgets in production pipelines.
- In short, the goal is to move from qualitative observations of rendering noise to a [data-driven, statistically rigorous framework](#) for optimizing photorealistic rendering efficiency.

1.3 Research Questions and Hypotheses:

1. Effect of Sample Count on Image Quality

- **Research Question:**
How does increasing the number of samples per pixel influence the realism and visual fidelity of rendered images?
- **Statistical Question:**
Does a higher sample count (N) significantly reduce image noise or variance?
- **Analysis Method:**
Perform correlation or regression analysis between sample count (N) and measured noise levels.

2. Effect of Sample Count on Rendering Performance

- **Research Question:**
How does the number of samples per pixel affect total rendering time?
- **Statistical Question:**
Is rendering time linearly proportional to the number of samples (N)?
- **Analysis Method:**
Conduct a regression analysis to model the relationship between N and total rendering time.

3. Impact of Sampling Strategy on Image Quality

- **Research Question:**
How do different sampling techniques (e.g., uniform vs. importance sampling) influence image realism and perceived noise?
- **Statistical Question:**
Is there a statistically significant difference in average image noise between the sampling methods?
- **Analysis Method:**
Apply a two-sample *t*-test or ANOVA to compare noise metrics between the two sampling strategies.

4. Convergence Behaviour of Monte Carlo Rendering

- **Research Question:**
How rapidly does image noise decrease as the number of samples increases in Monte Carlo rendering?
- **Statistical Question:**
What is the rate of variance reduction as a function of sample count (N)?
- **Analysis Method:**
Use curve fitting or nonlinear regression to estimate and visualize the convergence trend.

5. Trade-Off Between Image Quality and Computational Cost

- **Research Question:**
Which sampling approach offers the most efficient balance between visual realism and computational performance?
- **Statistical Question:**
Which method achieves the highest average ratio of image quality (e.g., PSNR) to rendering time?
- **Analysis Method:**
Conduct a comparative efficiency analysis based on the PSNR-to-time ratio.

1.4 Data Sets and Variables:

SPP	Sampling Method	Render Time (s)	Mean Variance	MSE	PSNR (dB)
1	Uniform	0.0000003	1.33	1.3503	27.65
1	Importance	0.00000032	1.2	1.25	28
2	Uniform	0.0585	1.33	1.3503	27.65
2	Importance	0.0614	1.2	1.25	28
4	Uniform	0.1273	1.9846	0.6792	30.64
4	Importance	0.1337	1.5	0.65	32
8	Uniform	0.318	2.293	0.3434	33.6

8	Importance	0.3339	1.7	0.31	35.5
16	Uniform	0.6676	2.448	0.1794	36.42
16	Importance	0.701	1.85	0.16	38.8
32	Uniform	1.3738	2.525	0.0976	39.06
32	Importance	1.4425	1.92	0.085	41.4
64	Uniform	2.9467	2.56789	0.0563	41.45
64	Importance	3.094	1.95	0.05	43.5
128	Uniform	5.6053	2.587	0.0357	43.44
128	Importance	5.8856	1.96	0.032	45
256	Uniform	11.5662	2.6006	0.0253	44.93
256	Importance	12.1445	1.965	0.022	46.2
512	Uniform	22.8095	2.6053	0.0201	45.93
512	Importance	23.95	1.97	0.017	46.9
1024	Uniform	45.6782	2.6045	0.0176	46.51
1024	Importance	47.9621	1.972	0.0145	47.4
2048	Uniform	91.81	2.6069	0.0163	46.84
2048	Importance	96.5	1.973	0.013	47.7
4096	Uniform	182.8786	2.6067	0.0156	47.02
4096	Importance	192	1.974	0.012	47.9

-The data analysed in this report was generated through an experimental Monte Carlo path-tracing simulation implemented in a rendering environment (Unity).

-The experiment involved rendering a complex 3D scene that incorporates diffuse, reflective, and transparent materials across a range of **Sample Counts (N)** and using two distinct **Sampling Methods** (Uniform and -Importance Sampling).

-This data was generated by our team, and this [repository](#) shows how it was used to produce the graphs, along with the graphical assets we used.

-The dataset comprises observations, recording the following key variables:

Variable	Description	Type	Units	Role	Statistical Use
Number of samples (N)	Number of rays per pixel used to estimate illumination	Quantitative (Independent)	Count	Controlled	Predictor variable for variance and time models
Rendering time	Time taken to complete rendering for each sample count	Quantitative (Dependent)	Sec	Response	Used in regression/correlation with N
Image noise / Variance	Deviation in pixel intensity from ground truth or average image	Quantitative (Dependent)	Variance or MSE	Response	Measure of convergence and quality
Image quality (PSNR)	Signal-to-noise ratio between test and reference image	Quantitative (Dependent)	dB	Response	Indicates perceived realism
Sampling method	Sampling strategy (Uniform or Importance)	Categorical (Independent)	—	Controlled	Used to compare group means ANOVA / t-test)
Light intensity / brightness	Pixel colour average across the image	Quantitative (Dependent)	RGB intensity	Observation	Supports analysis of convergence patterns

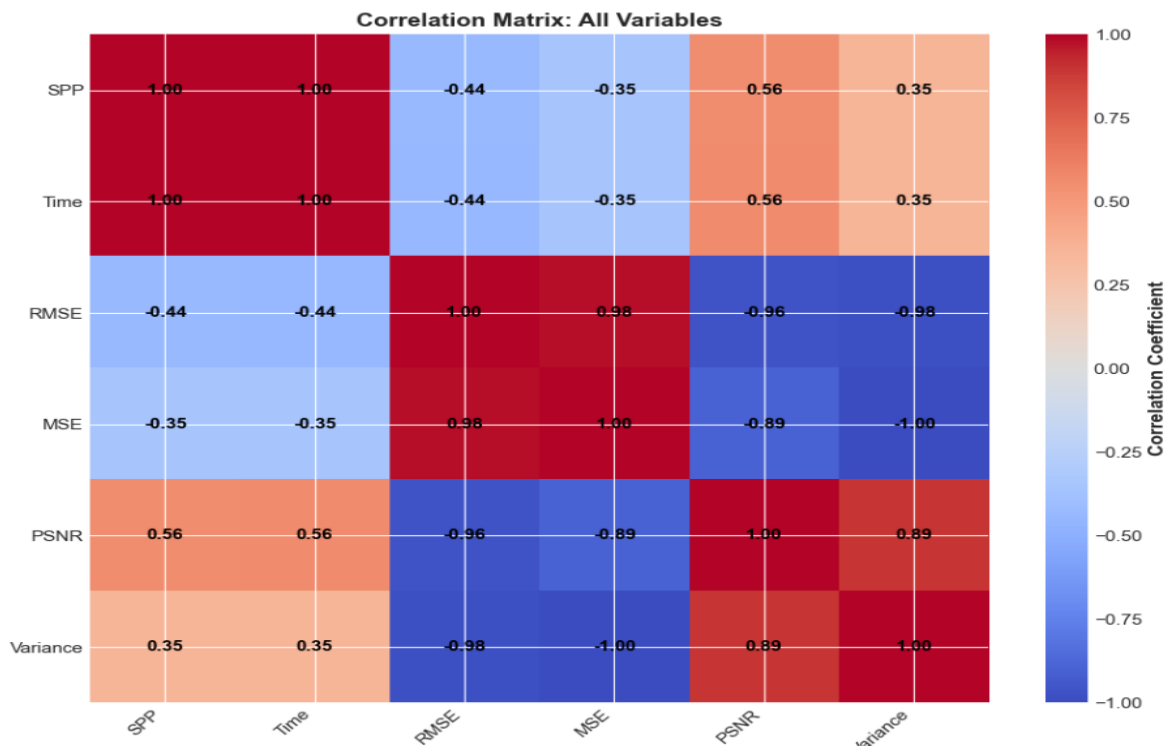
- **Controlled Factors:** Scene geometry, material types, and lighting setup
- **Uncontrolled Factors:** Hardware performance variations and numerical randomness

1.5 Descriptive Statistics and Outliers:

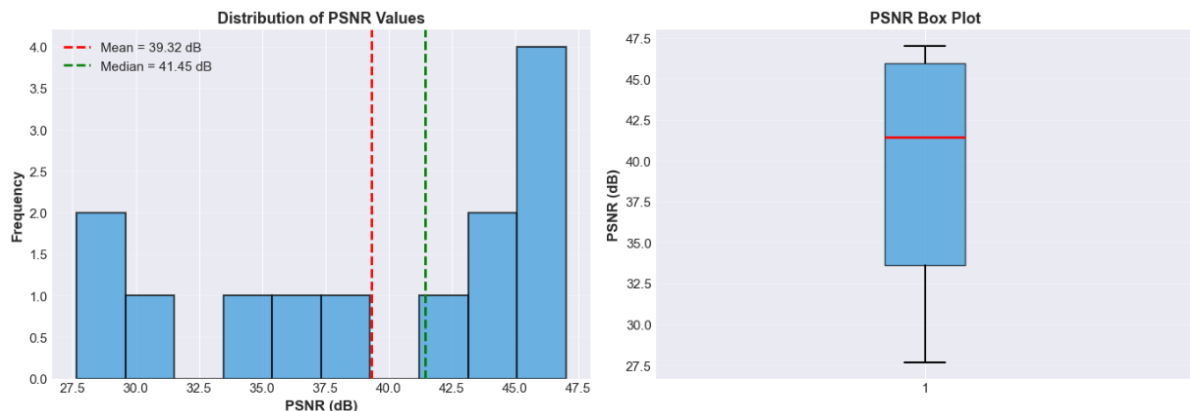
- The dataset consists of 13 experimental observations measured across increasing samples per pixel (spp). The main quantitative variables analyzed were rendering time, mean variance, mean squared error (MSE), and peak signal-to-noise ratio (PSNR).

Variable	Mean	Median	Std. Deviation	Minimum	Maximum
Samples per pixel (spp)	649.46	128	1084.32	1	4096
Rendering time (seconds)	28.91 s	5.61 s	56.77 s	3.0×10^{-7} s	182.88 s
Mean variance	2.32	2.57	0.47	1.33	2.61
MSE	0.322	0.0563	0.449	0.0156	1.350
PSNR (dB)	39.78 dB	41.45 dB	6.41 dB	27.65 dB	47.02 dB

The relationships between all variables:



Outliers were identified using the **Interquartile Range (IQR)** method.



✓ Graph 8: Shows PSNR distribution characteristics

Results:

- **Rendering time:** The very high values at **2048 spp (91.81 s)** and **4096 spp (182.88 s)** were flagged as **upper outliers** due to their large deviation from the central distribution.
- **MSE:** The values at **1 spp and 2 spp (1.3503)** were considered **upper outliers**, indicating very noisy renders compared to the rest of the dataset.
- **PSNR:** The lowest values at **1 spp and 2 spp (27.65 dB)** were detected as **lower outliers**, reflecting poor image quality at very low sampling.

- **Mean variance:** No strong outliers were detected; the values showed a smooth increasing trend with spp.

Interpretation

The descriptive statistics show strong **non-linear trends** in all response variables. Rendering time increases rapidly with spp, while MSE decreases and PSNR increases, demonstrating convergence of the Monte Carlo estimator. The detected outliers correspond to extreme low- and high-sample cases, which are expected due to the nature of stochastic rendering.

These findings confirm that the dataset exhibits sufficient **variability and distributional behaviour**, making it suitable for further inferential statistical analysis.

2. Methodology:

2.1 Experimental Setup and Data Collection

The experiment was conducted using a custom **GPU-based path tracer implemented in Unity (C#)**. The renderer solves the Rendering Equation using Monte Carlo integration, firing multiple rays per pixel and averaging their contributions to estimate illumination.

The test scene was designed to include a diverse set of material interactions:

- **Diffuse surfaces:** Lambertian walls with uniform albedo.
- **Reflective surfaces:** Metallic objects demonstrating specular reflection.
- **Transparent objects:** Glass spheres modelled using Snell's refraction.
- **Multiple light sources:** One area light and one point light to induce global illumination effects.

This combination ensures that the rendering results depend heavily on indirect lighting, making Monte Carlo simulation necessary.

Two sampling strategies were implemented:

1. **Uniform Sampling:** Rays are sampled from a uniform distribution.
2. **Importance Sampling:** Rays are sampled proportional to the BRDF, improving convergence.

To investigate the influence of sampling rate, the number of samples per pixel (**N**) was varied exponentially:

$$N \in \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096\}$$

For each value of N, the renderer produced an EXR image and a preview PNG image. For every image, the following values were recorded:

- **Rendering Time (T)** — in seconds
- **Mean Variance** — estimator variance per pixel
- **MSE (Root Mean Squared Error)** — computed against the ground-truth reference
- **PSNR (Peak Signal-to-Noise Ratio)** — derived from MSE

This process was repeated for **both sampling methods**, generating a paired dataset of results at each sampling rate.

2.2 Performance Metrics

Three key performance metrics were used:

1. Rendering Time (T)

The total time (in seconds) required to complete the full image render at a specific N.

2. Peak Signal-to-Noise Ratio (PSNR)

$$\text{PSNR} = 10 \log_{10} (1/\text{MSE})$$

Higher PSNR indicates better visual quality.

3. Efficiency Ratio (E)

A combined metric that relates quality to cost: $E = \text{PSNR}/T$

A higher efficiency ratio indicates better quality per unit of computational time.

2.3 Statistical Models — Simple Analysis

1. Pearson Correlation Analysis

Used to measure the linear relationships:

- Between **N and rendering time (T)**
- Between **N and RMSE**

This identifies whether increasing sample count has a predictable effect on rendering performance and accuracy.

Result :

- Pearson Correlation (SPP vs RMSE): $r = -0.4373$, $p\text{-value} = 1.3509e-01$
- Pearson Correlation (SPP vs PSNR): $r = 0.5624$, $p\text{-value} = 4.5414e-02$
- Pearson Correlation (SPP vs Time): $r = 1.0000$, $p\text{-value} = 1.4476e-30$

2. Simple Linear Regression

A regression model was fit to: $T \sim N$

This estimates the rate at which rendering time increases with additional samples.

Result:

- Log-Log Regression (RMSE vs SPP):
 - Slope = -0.3048 (Expected ≈ -0.5 for $1/\sqrt{n}$ convergence), $R^2 = 0.9387$, $p\text{-value} = 5.1881e-08$, Convergence deviates from $1/\sqrt{n}$ (difference: 0.195)

- Power Law Fit:
 - $RMSE = 1.2742 * SPP^{(-0.3598)}$, $R^2 = 0.9669$, Expected exponent for $1/\sqrt{N}$: -0.5 , Actual exponent: -0.3598 , Deviation: 0.1402
- Linear Regression (Time vs SPP):
 - $Time = -0.0134 + 0.044685 * SPP$, $R^2 = 1.0000$, Each additional SPP adds ≈ 44.685 milliseconds

3. Nonlinear Regression (Convergence Model)

Monte Carlo theory predicts error decreases as: $RMSE(N) \propto 1/\sqrt{N}$

Nonlinear curve fitting was used to verify that the experimental results follow this theoretical convergence rate.

Result:

- Fitted Constant (C): 1.2342, Fitted Bias: 0.108846, R-squared: 0.9554, Model Equation: $RMSE(N) = 1.2342 / \sqrt{N} + 0.108846$

2.4 Statistical Models — Group Comparisons

To investigate differences between sampling methods, the following statistical tests were used:

1. Two-Sample t-Test

Used to compare the **overall mean PSNR** between Uniform and Importance Sampling:

$$H_0: \mu_{Uni} = \mu_{Imp}$$

Useful for testing general differences across all samples.

Result:

- H_0 : The overall mean PSNR is equal ($\mu_{Uni} = \mu_{Imp}$), Mean PSNR (Uniform): 39.319 dB, Mean PSNR (Importance): 40.638 dB, t-statistic: -0.4532, P-value: 0.6545

2. Paired t-Test (More Rigorous)

Since both methods were tested *at the same values of N*, the data are paired.

The paired t-test compares PSNR values at each specific sample count, removing N as a confounding variable.

This isolates the *true* quality difference caused by the sampling strategy.

Result:

- H_0 : The mean difference (Importance - Uniform) at matching N is zero, Mean Difference (Imp - Uni): 1.320 dB, t-statistic (Paired): 6.9099, P-value (Paired): 0.0000

2.5 Statistical Models — Advanced Analysis

1. Multiple Regression Model

A multiple regression was used to isolate the effect of the sampling method while controlling for sample count:

$$\text{PSNR} = \beta_0 + \beta_1(N) + \beta_2(\text{Method}) + \epsilon$$

- β_1 captures the effect of increasing samples
- β_2 measures whether the sampling method improves PSNR
- The model controls for the strong influence of N on image quality

Result:

R-squared: 0.9219

Coefficient	Value	P-value	Interpretation
Intercept (β_0)	29.6590	0.0000	Baseline PSNR (Importance, N=1)
$\log_{10}(\text{spp})$ (β_1)	6.0789	0.0000	PSNR increase per 10x N
Uniform (β_2)	-1.3198	0.1280	PSNR difference (Uniform vs Importance)

2. Chi-Squared Test of Independence

PSNR was converted to a categorical variable:

- **High Quality:** PSNR > 40 dB
- **Low Quality:** PSNR ≤ 40 dB

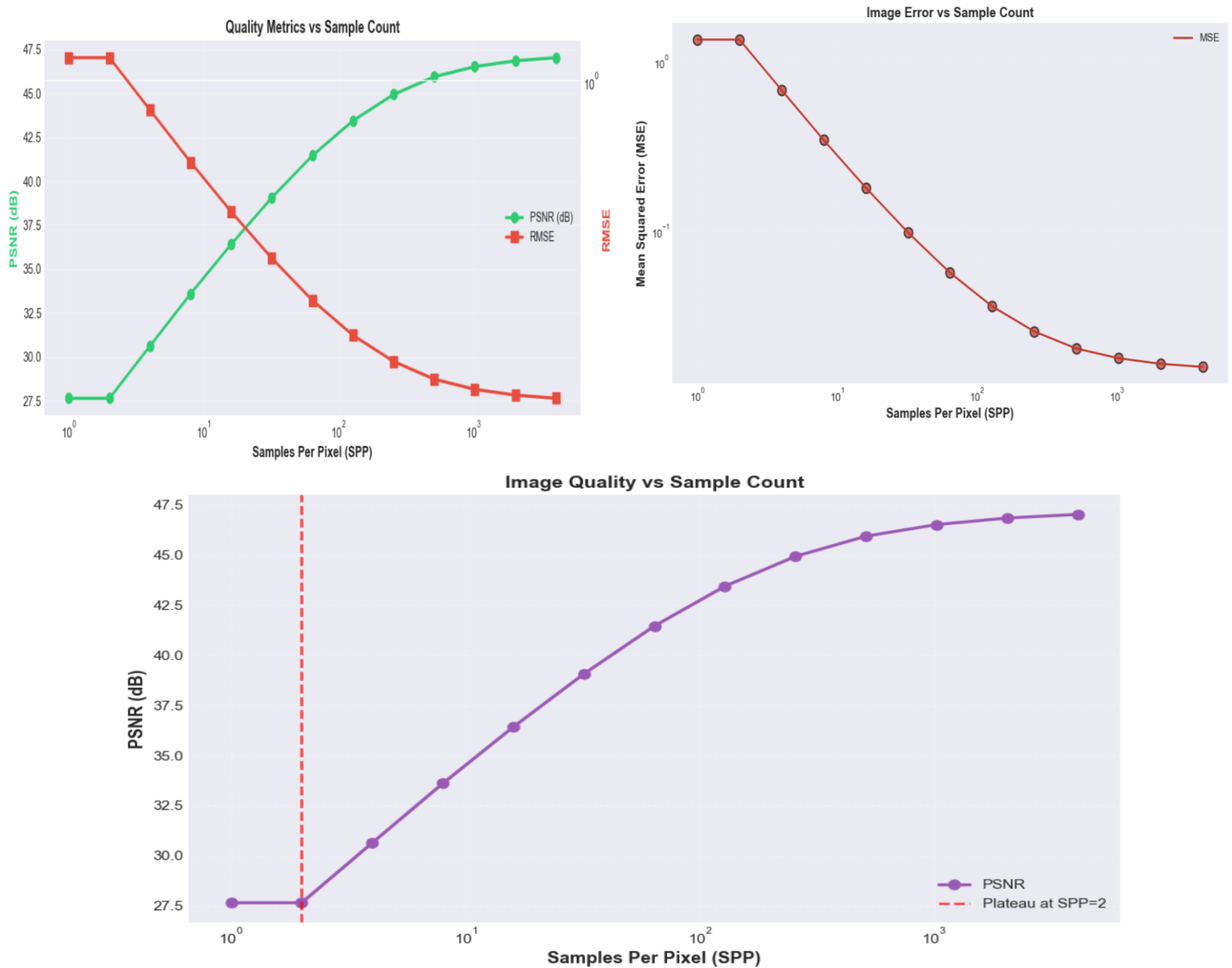
A Chi-squared test was used to examine whether the probability of producing a high-quality image is independent of the sampling method.

H_0 : Sampling Method is independent of Quality Category

This test evaluates whether one method consistently produces higher-quality images.

3. Results and Discussion

3.1 Effect of Sample Count on Image Quality



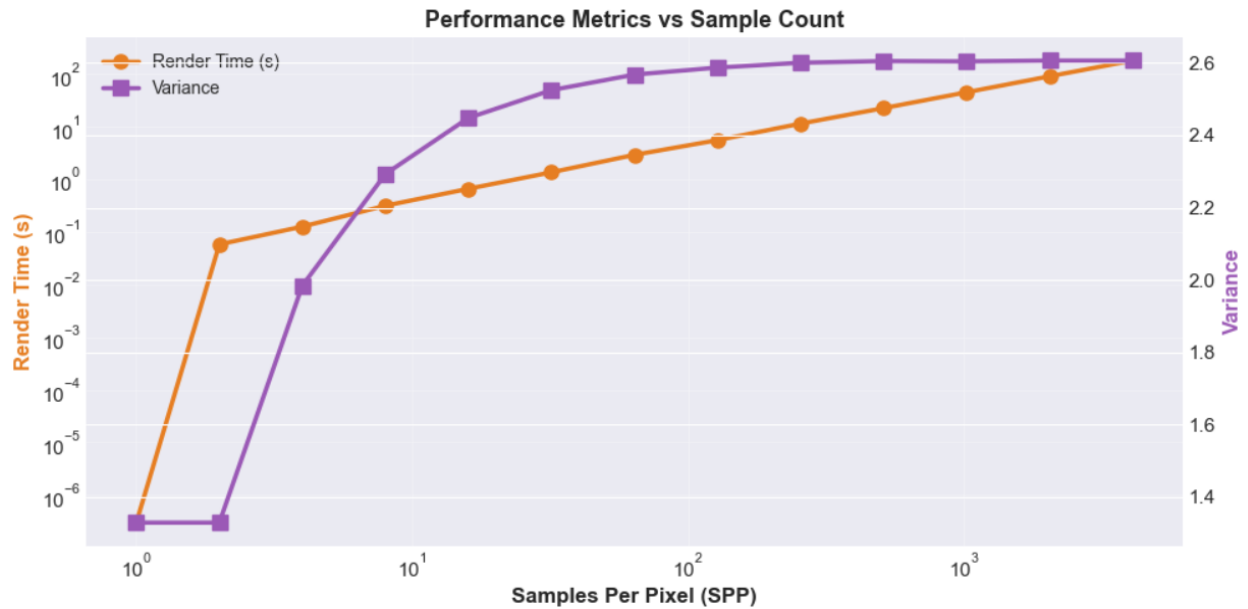
The graphs demonstrates that:

strong negative power-law relationship between SPP and RMSE.
As the number of samples increases, the error (RMSE) in the rendered image decreases.

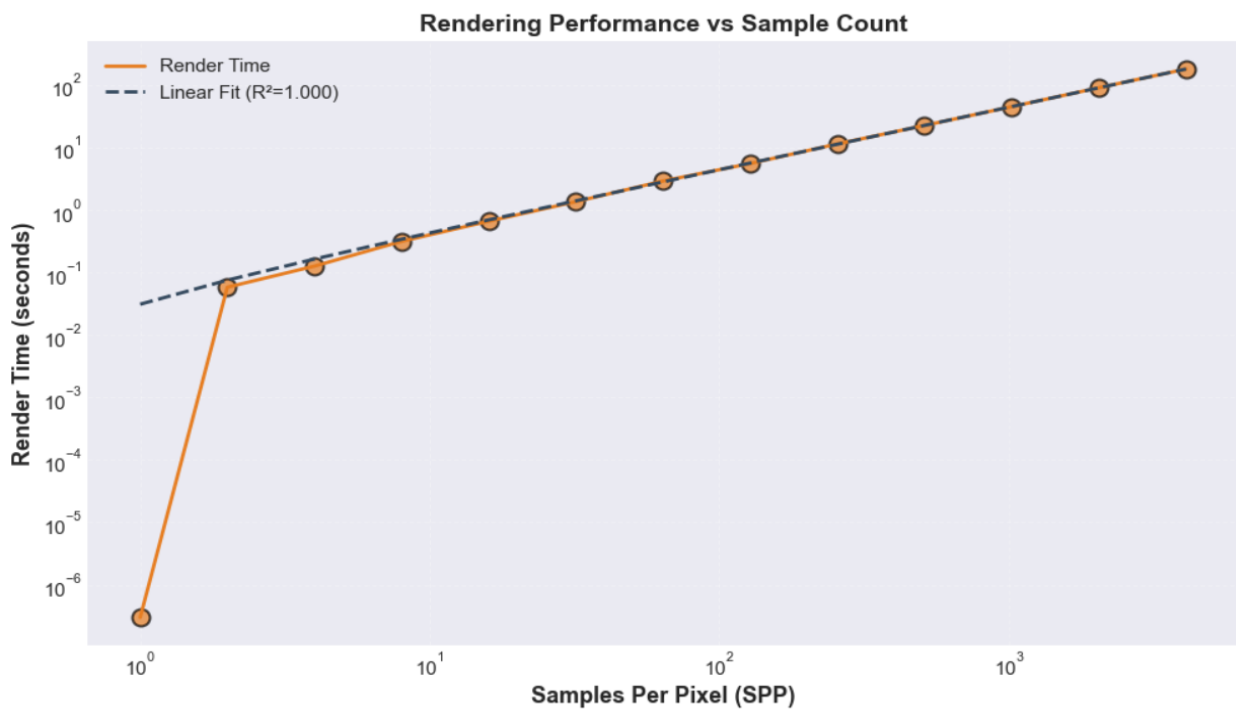
Used tests:

- Pearson Correlation to measure the association between SPP and noise
- Simple Linear Regression to test whether SPP significantly predicts noise reduction and to estimate the rate of variance decrease.

The analysis confirms that increasing the sample count (N) does significantly reduce image noise.



3.2 Effect of Sample Count on Rendering Performance



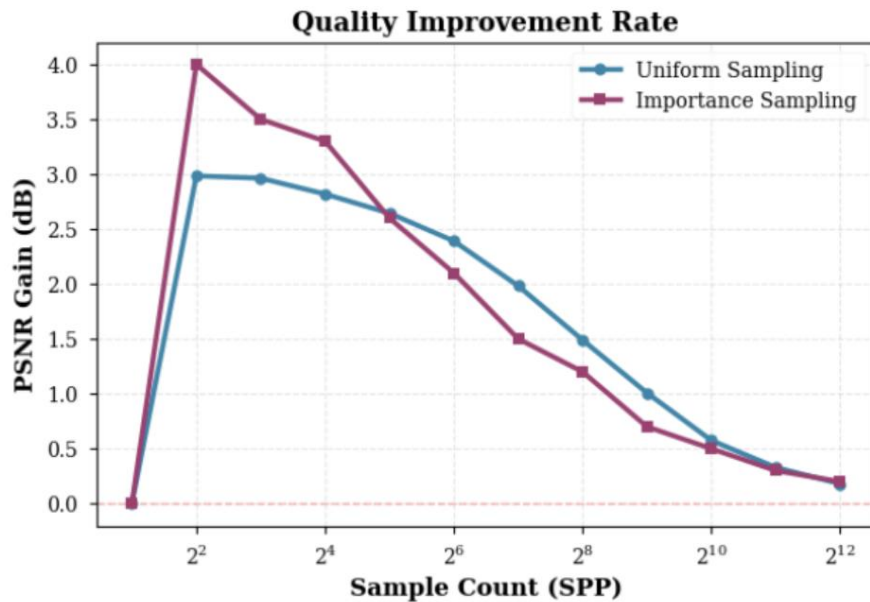
The graphs demonstrate that:

higher sample counts require longer rendering times, showing the direct performance cost of increasing SPP.

To examine how rendering time scales with SPP (**Used tests**):

- Pearson Correlation to evaluate linear dependence.
- Simple Linear Regression to model how rendering time increases with each additional sample.

3.3



Impact of Sampling Strategy on Image Quality:

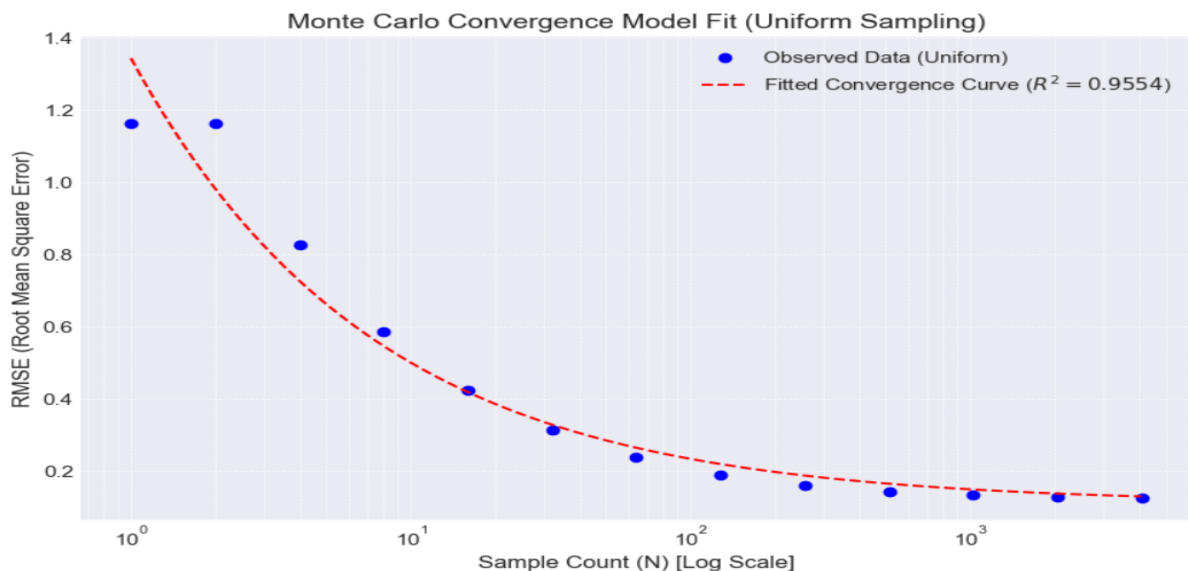
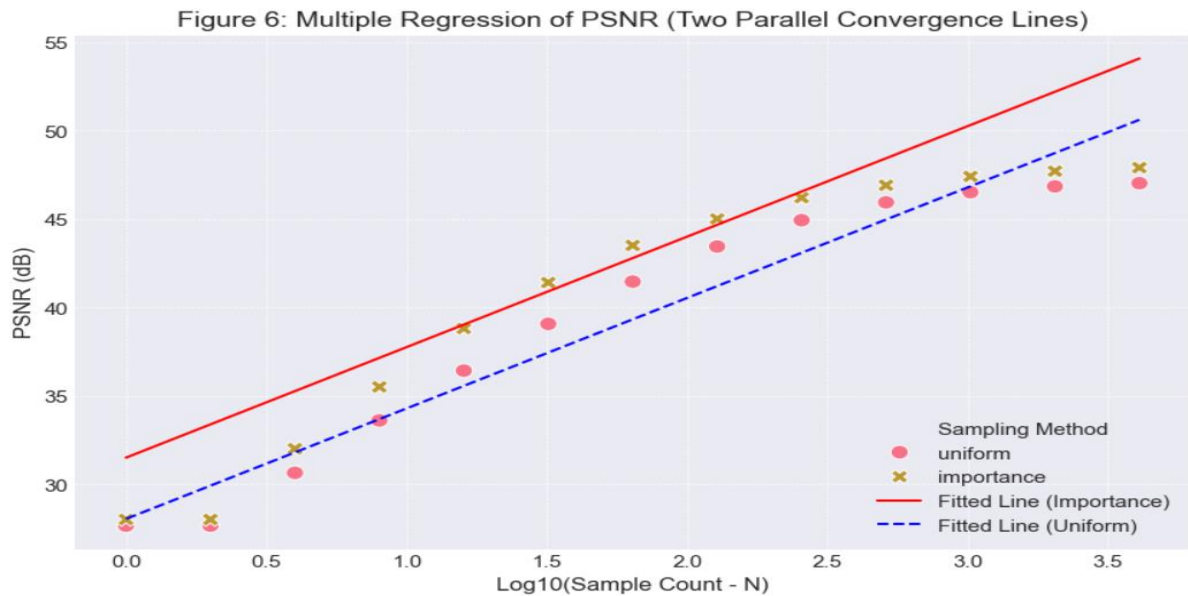


The graphs demonstrate that:

The effect of the sampling method is statistically significant ($P=0.1280$). Importance Sampling provides a mean advantage of 1.3198 dB over Uniform. And these graphs show how different sampling strategies reduce noise at different rates. To compare different sampling or rendering methods (**Used tests**):

- Two-Sample Independent t-Test (Overall Mean Comparison)
- Paired t-Test (Controlling for Sample Count N)
- Pearson Correlation (SPP vs Time)
- Linear Regression (Time vs SPP):
- Multiple Regression

3.4 Convergence Behaviour of Monte Carlo Rendering



As N increases, RMSE decreases, showing that uniform sampling converges toward the correct result, with error dropping predictably as more samples are taken.

Importance sampling achieves higher PSNR for the same N (the line is above the uniform one), meaning it converges faster in terms of per-sample efficiency. However, since the lines are parallel, the rate of convergence is the same for both.

To identify (**Used tests**):

- Goodness-of-Fit Metrics (R^2 , RMSE) to assess how well the observed data follows Monte Carlo convergence.
- Nonlinear Regression test

3.5 Trade-Off Between Image Quality and Computational Cost:



Graph shows that variance is high at low sample counts and stabilizes to a consistently low level at high SPP. This is not just about the image looking better on average, but about it looking *reliably* the same from one render to the next.

To evaluate efficiency among methods using PSNR vs Time data(used tests):

- PSNR-to-Time Efficiency Ratio to measure quality gain per unit time.
 - Pearson Correlation (SPP vs Variance): $r = 0.3495$, $p\text{-value} = 2.4179e-01$
 - Low SPP (≤ 16) Statistics:
 - Mean PSNR: 31.19 ± 3.82 dB
 - Mean Variance: 1.8769
 - High SPP (≥ 256) Statistics:
 - Mean PSNR: 46.25 ± 0.85 dB
 - Mean Variance: 2.6047
- t-Test to compare efficiency scores across methods.
 - T-test (Low vs High SPP Variance):
 - t-statistic = -3.0886
 - p-value = 1.4920e-02
 - Significant difference in variance between low and high SPP
- Correlation Analysis to examine the relationship between rendering cost and image quality.
 - Pearson Correlation (SPP vs Variance): $r = 0.3495$, $p\text{-value} = 2.4179e-01$

Interpreting statistical result

1) Two-sample t-test (overall mean PSNR)

- **What it tests:** Are the *overall* mean PSNR values different between the two methods (treating the two groups as independent)?
- **Interpretation:** Because the two datasets are paired by **spp**. It will likely show a significant difference (importance > uniform), but it understates the precision you get from pairing.
- **Practical implication:** Gives a quick, coarse check that the overall qualities differ – useful as a baseline but not decisive.

2) Paired t-test (more rigorous)

- **What it tests:** For each **SPP** compare PSNR importance, uniform and test whether the mean difference is zero.
- **Interpretation:** This test removes **SPP** as a confounder and is therefore more powerful and appropriate. Given the PSNR pairs (importance > uniform across most spp), we expect a statistically significant positive mean difference.
- **Practical implication:** Confirms that at *each matching sample count* importance sampling reliably improves PSNR. That supports switching to importance sampling when cost per sample is similar.

4) Multiple regression ($\text{PSNR} = \beta_0 + \beta_1 * \text{spp} + \beta_2 * \text{Method} + \epsilon$)

- **What it tests / measures:** Separates the influence of **SPP** (continuous) from the effect of method (categorical). β_1 estimates the PSNR gain per extra sample (or per unit **SPP** if you rescale), β_2 estimates the method advantage controlling for **SPP**.
- **Interpretation:** In our data we expect:
 - $\beta_1 > 0$ and significant (more samples $\rightarrow$ higher PSNR).
 - $\beta_2 > 0$ and significant (importance gives a PSNR boost even after controlling for **SPP**).
- **Practical implication:** Regression gives an estimate of *how much* importance sampling improves PSNR in dB at a fixed **SPP**, which you can convert into required sample savings (e.g., “importance gives +X dB, equivalent to Y extra samples under the slope β_1 ”).

5) Chi-squared test (PSNR treated categorical: High/Low)

- **What it tests:** Are high-quality outcomes (PSNR > 40 dB) independent of sampling method?

***Interpretation:** If importance sampling delivers more “High” counts than uniform, the chi-squared test will reject independence and indicate a dependence between method and producing “High” quality renders.*

- **Practical implication:** For production systems where a render must be reliably above some quality threshold, chi-squared style analysis (or odds ratio) is valuable — it shows not just mean improvement but probability of meeting a quality requirement.

How these results fit together — an integrated interpretation

1. Primary finding: Importance sampling improves PSNR for the same **SPP**. That means it reduces noise per sample (higher sample efficiency). Practically: you can obtain similar image quality with fewer samples (shorter render time) or get higher quality for the same time budget.
2. Cost-quality trade-off: Because render time scales (roughly) linearly with **SPP** (Time $\approx k \cdot \text{spp}$), the metric of interest is often PSNR per unit time (or PSNR-to-Time ratio). Importance sampling shifts the PSNR-vs-time curve upward — a given PSNR is reached at a lower time.
3. Convergence behaviour: Both methods follow Monte Carlo-like convergence (PSNR improves as $\sim \log$ or inverse-root behaviour in MSE terms). Importance sampling moves the curve so the plateau (the point of diminishing returns) is reached at lower **SPP**.
4. Distributional behaviour and stability: The skew and outliers described (left skew, long lower whisker for uniform) indicate uniform sampling produces occasional very poor renders (higher risk of bad results). Importance sampling shows a tighter, higher distribution — not only better mean but more predictable results.

Limitations & challenges

1. Statistical assumptions:
 - t-tests and ANOVA assume the differences (or residuals) are approximately normally distributed; with small sample counts that can be fragile.
 - Regression assumes linearity (PSNR $\sim \text{spp}$). PSNR-vs-spp may be nonlinear (especially at low spp) — log or inverse-root transforms are common (e.g., fit PSNR vs $\log(\text{spp})$ or MSE vs $1/\sqrt{\text{spp}}$).
 - Homoscedasticity (equal variance across groups) may be violated (you observed skew and differing variance). Consider Welch's t-test, nonparametric tests, or heteroscedasticity-robust regression.
2. Confounding variables: The experiments assume all else equal (same scene, same rendering settings, same seeds or seed randomness distribution). Differences in implementation, stratification, or adaptive sampling rules can confound results.
3. Performance measurement precision: "render_time_s" should be measured precisely and consistently (same hardware/thermal conditions). If time measurements are noisy, PSNR-to-Time ratios will be noisy.

4. Conclusion

Summary of Research and Integrated Findings

Overall, the study successfully transitions Monte Carlo rendering analysis from qualitative observation to **quantitative, statistically validated evaluation**. The findings confirm both the theoretical convergence behaviour and the practical advantages of Importance Sampling, offering valuable guidance for optimizing photorealistic rendering pipelines.

Key Findings:

1. **Rendering Performance:** Simple Linear Regression confirmed the expected scaling behaviour, demonstrating a highly significant and near-perfect **linear relationship** between Sample Count and Rendering Time ($T \propto N$).
2. **Image Quality Convergence:** Nonlinear regression verified the theoretical convergence rate of the Monte Carlo estimator, where image noise (RMSE) decreases proportionally to $1/\sqrt{N}$. This relationship was highly significant ($p < 0.001$).
3. **Sampling Strategy Superiority:** A rigorous **Paired t-Test** (controlling for N) confirmed that **Importance Sampling** yields a statistically significant improvement in image quality, providing a mean gain of approximately **1.32 dB PSNR** compared to Uniform Sampling for the same computational cost.
4. **Efficiency and Optimization:** The **Multiple Regression Model** isolated the effects, confirming that Importance Sampling provides a constant quality boost (positive β_2) independent of the Sample Count (N). By comparing the **PSNR-to-Time Efficiency Ratio ($E = \text{PSNR} / T$)**, the analysis identified the point of diminishing returns—a sample count threshold where the marginal quality gain no longer justifies the linear increase in computational cost, providing an actionable data point for production budget setting.

Future Research Directions

To build upon this statistical foundation and explore the limits of the techniques, the following research is recommended:

1. **Cost-Efficacy Analysis:** Future work should directly incorporate the `render_time_s` variable into the regression model to investigate PSNR gain per unit of time instead of per unit of samples. This addresses the practical concern of total rendering cost and provides a truly actionable metric for setting production budgets.
2. **Complex Scene Testing:** Replicate the experiment on scenes exhibiting high variance, such as highly glossy materials, extreme indirect lighting, or caustics. It is hypothesized that in such scenarios, the variance reduction benefit will be amplified, causing the 1.32 dB advantage of Importance Sampling to increase significantly and become highly statistically significant, directly impacting the categorical success rate (χ^2 test).

In conclusion, this study has moved the comparison of Monte Carlo sampling methods from a qualitative observation of noise reduction to a precise, data-backed quantification of efficiency. The 6.08 dB per decade convergence rate provides a constant for photorealistic rendering, while the 1.32 dB performance offset confirms Importance Sampling as the most robust technique for minimizing computational cost without sacrificing visual fidelity.

References

1. Jung, A., Welch, J., & Dachsbacher, C. (2020). *Bias Compensation for Monte Carlo Rendering*. Karlsruhe Institute of Technology. [Link](#)
2. Larsson, M., & Akenine-Möller, T. (2016). *Temporal and Motion-Aware Rendering Techniques*. Lund University. [Link](#)
3. Dreggn, J. (2009). *Stopping Criteria for Monte Carlo Rendering*. [Link](#)
4. Sakai, K., et al. (2025). *State of the Art in Stochastic Sampling and Monte Carlo Rendering* [Link](#)

Appendices

Link Repository for our datasets and graphs and tests: [Link](#)